

Test Report — Fastener Retention Under Vibration

Test Laboratory · 2025-03 · Issue 1

1. Test setup

This report records a fastener retention assessment performed on the housing interface under vibration loading. The assessment was requested through the standard booking route and performed on rig 2.

Field	Entry
Report	TL-RPT-2025-044
Rig	Rig 2
Specimens	24
Fastener	M8 housing bolt, as supplied
Preparation	Tightened to the specified value, no defined sequence

Specimens were prepared to the tightening method in the instruction current at the time of preparation. No tightening sequence was applied, as none was specified. Fasteners were fitted dry.

Instrumentation and fixture details are recorded in the test file. Residual torque was measured on each fastener at the intervals stated in section 2.

2. Results

Residual torque was measured at four intervals through the vibration sequence. Values are stated as a percentage of the value applied at preparation.

Interval	Mean residual	Lowest specimen
2 hours	94%	88%
8 hours	81%	69%
24 hours	72%	54%
48 hours	68%	47%

Residual torque fell across every specimen. Nineteen of twenty-four specimens fell below the specified value before the 24-hour interval. No specimen recovered residual torque at any subsequent interval.

A second set of twelve specimens was prepared with the fasteners tightened in a diagonal order and rechecked thirty minutes after initial tightening. These specimens were run through the identical sequence.

Interval	Mean residual, second set	Lowest specimen
2 hours	98%	96%
8 hours	96%	93%
24 hours	95%	91%
48 hours	94%	90%

No specimen in the second set fell below the specified value at any interval.

3. Observations

The condition observed in the first set is reproducible under the sequence applied and was observed in every specimen of that set.

The second set differed from the first in preparation method only. Specimen material, fastener supply, fixture and vibration sequence were identical across both sets.

Variable	First set	Second set
Tightening order	None specified	Diagonal
Recheck after tightening	None	At thirty minutes
Specimens below specified value at 48 hours	22 of 24	0 of 12

This report records the measurements taken. Determination of any action arising is outside the scope of the test laboratory and is not made here.